

Daniel Waxman

Basis Research Institute
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EDUCATION

- Ph.D. Electrical Engineering, *December 2025*
Stony Brook University
Thesis: *Sequential Bayesian Learning and Ensembles for Online Inference and Experimental Design*
Advisor: Petar M. Djurić
- B.S. Mathematics & Applied Mathematics and Statistics (*cum laude*), May 2021
Stony Brook University

EMPLOYMENT & ACADEMIC APPOINTMENTS

- 01/26 – Postdoctoral Research Scientist
Present *Basis Research Institute*, Cambridge, MA.
Advised by Matt Levine at Basis and Youssef Marzouk at MIT on Bayesian inference and causality in dynamical systems.
- 06/25 – Research Intern
01/26 *Basis Research Institute*, New York, NY.
Research in causal inference, probabilistic programming, and applied Bayesian machine learning with Rafal Urbaniak and Jack Feser.

RESEARCH INTERESTS

- Bayesian Machine Learning — particularly problems in dynamical systems and online settings
- Ensembling Methods — particularly Bayesian-based ensembling rules and online ensembling
- Causal Inference — particularly causal discovery and causality for dynamical systems

PUBLICATIONS

* denotes equal contribution

Journal Papers

- 2026 **D. Waxman**, F. Llorente, P. M. Djurić, “Sequential Inference with Gaussian Processes: A Signal Processing Perspective.” *Submitted; Invited for Submission to IEEE Signal Processing Magazine*.
- 2026 **D. Waxman**, F. Llorente, P. M. Djurić. “Bayesian Ensembles: Insights from Online Optimization and Empirical Bayes.” *Submitted*.
[\[arXiv\]](#) [\[code\]](#)
- 2025 **D. Waxman**, F. Llorente, K. Lamer, P. M. Djurić, “Designing an Optimal Sensor Network via Minimizing Information Loss.” *Accepted to Bayesian Analysis, 2026*.
[\[arXiv\]](#) [\[code\]](#)

- 2024 **D. Waxman**, P. M. Djurić. “Dynamic Online Ensembles of Basis Expansions.” *Transactions on Machine Learning Research (TMLR)*, 2024.
[\[OpenReview\]](#) [\[arXiv\]](#) [\[code\]](#)
- 2024 **D. Waxman**, K. Butler, P. M. Djurić. “DAGMA-DCE: Interpretable, Non-Parametric Differentiable Causal Discovery.” *IEEE Open Journal of Signal Processing*, vol. 5, pp. 393-401, 2024.
[\[IEEE Xplore\]](#) [\[arXiv\]](#) [\[code\]](#)

Conference Papers

- 2026 R. Urbaniak, S. Witty, **D. Waxman**, A. Zane, J. Feser, P. Garg, D. Lehe, E. Bingham. “A Computationally Feasible Framework for Causal Probabilistic Explanation.” Submitted, 2026.
- 2026 F. Llorente, **D. Waxman**, S. Jantre, N. Urban, S. Minkoff. “Robust, Online, and Adaptive Decentralized Gaussian Processes.” Submitted, 2026.
[\[arXiv\]](#)
- 2025 **D. Waxman**, P. M. Djurić. “Non-Stationary Causal Learning via Hierarchical Modeling.” *Proceedings of the 2025 Asilomar Conference on Signals, Systems, and Computers*, 2025.
- 2025 F. Llorente*, **D. Waxman***, P. M. Djurić. “Decentralized Online Ensembles of Gaussian Processes for Multi-Agent Systems.” In *Proceedings of the 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
[\[IEEE Xplore\]](#) [\[arXiv\]](#) [\[code\]](#)
- 2024 K. Butler*, **D. Waxman***, P. M. Djurić. “Tangent Space Causal Inference: Leveraging Vector Fields for Causal Discovery in Dynamical Systems.” In *Advances in Neural Information Processing Systems*, 37, 120078-120102, 2024.
[\[OpenReview\]](#) [\[arXiv\]](#) [\[code\]](#)
- 2024 **D. Waxman**, P. M. Djurić. “A Gaussian Process-based Streaming Algorithm for Prediction of Time Series With Regimes and Outliers.” In *Proceedings of the 27th International Conference on Information Fusion (FUSION)*, Venice, Italy, 2024.
[\[IEEE Xplore\]](#) [\[arXiv\]](#) [\[code\]](#)
- 2023 M. Ajirak, **D. Waxman**, F. Llorente, P. M. Djurić. “Fusion of Gaussian Process Predictions with Monte Carlo Sampling.” *Proceedings of the 57th Asilomar Conference on Signals, Systems, and Computers* (pp. 1367-1371), 2023.
[\[IEEE Xplore\]](#) [\[arXiv\]](#)
- 2023 Y. Liu, C. Cui, **D. Waxman**, K. Butler, P. M. Djurić. “Detecting Confounders in Multivariate Time Series Using Strength of Causation.” In *Proceedings of the 31st European Signal Processing Conference (EUSIPCO)*, pp. 1400-1404, 2023.
[\[IEEE Xplore\]](#) [\[pdf\]](#) [\[video\]](#)

Professional Meeting Abstracts

- 2026 K. Lamer, E. V. Davis, Z. J. Mages, P. Joshi, N. Rojas, E. R. Vivoni, Z. Wang, Z. Zhu, **D. Waxman**. “Multisensor Observations and Meter-Scale Modeling of Urban Park Cooling Dynamics for Digital Twin Integration.” *106th AMS Annual Meeting*.
- 2024 D. Chernoff, **D. Waxman**, G. Brite, L. Langman, C Rabeno, L. Czerwinka, P. M. Djurić. “The Development of an Automated Algorithm to Identify and Manage Post-thyroidectomy Hypocalcemia.” *American Head and Neck Society 2024 General Meeting*.

RESEARCH TALKS

- 2026 SIAM Uncertainty Quantification '26
Minisymposium on “Sequential Inference for Dynamical Systems”
Talk Title: “GP4SP: Sequential Gaussian Process Inference for Signal Processing”
- 2026 Queens College CUNY Computer Science Colloquium
Talk Title: “Online Bayesian Learning and Ensembles”
- 2024 2024 SIAM New York-New Jersey-Pennsylvania Section Conference
Talk Title: “Optimizing Observation Locations via Minimizing Information Loss”
- 2023 Acoustics Research Institute of the Austrian Academy of Sciences
Talk Title: “Causal Discovery via Quantifying Influences” [\[link\]](#) [\[slides\]](#)
- 2023 Bellairs Workshop on Machine Learning and Statistical Signal Processing for Data on Graphs
Talk Title: “Bayesian Combination”

TEACHING

As Primary Instructor

- Fall 2025 Understanding Machine Learning (ESE 188 @ SBU)
Spr 2025 Understanding Machine Learning (ESE 188 @ SBU)

As Teaching Assistant

- Sum 2022 Random Signals and Systems (ESE 306 @ SBU)
Spr 2022 Random Signals and Systems (ESE 306 @ SBU)
Fall 2021 Programming Fundamentals (ESE 124 @ SBU)

OUTREACH & SERVICE

Reviewing

- Journals Transactions on Machine Learning Research (TMLR) [2024, 2025]
Journal of Open Source Software (JOSS) [2024, 2025, 2026]
EURASIP Journal on Advances in Signal Processing [2023]
- Conf. NeurIPS [2025; **Top Reviewer** 2025]
ICLR [2025, 2026]
ICML [2025, 2026]
EUSIPCO [2024, 2025]
NeurIPS Workshops [2024]
IEEE SSP [2025]
Asilomar [2025]
AAAI [2026]

University Service

- 2021–25 Stony Brook University Graduate Student Organization
Primary Senator, Dept. of Electrical & Computer Engineering
Graduate Student Representative for Graduate Council
Reviewer for Distinguished Travel Award

- 2022–23 SBU Strategic Planning Committee [\[link\]](#)
Member of Strategic Planning Committee; helped draft vision statement for Project REACH

Expository Talks

- 2024 SBU ECE Honors Seminar
Ensembling and Fusion: Perspectives and Solutions from Signal Processing
- 2022 Stony Brook Math Club
Making Markov Chains with Metropolis [\[video\]](#)

Organizing

- 2026 Minisymposium on “Sequential Inference for Dynamical Systems”
Co-organized with Fernando Llorente-Fernandez
SIAM UQ 2026, Minneapolis, MN.
- 2025 Special Session on “Advances in Causal Inference: Theory and Applications”
Co-organized with Petar M. Djurić
Asilomar 2025, Pacific Grove, CA

Mentoring

- 2022–25 CUNY Directed Reading Program
Mike Prebil (*Gaussian Processes and Bayesian Optimization*)
Edgar Cuapio Diaz (*Gaussian Processes and Bayesian Optimization*)
Jonathan Jaimangal (*Hamiltonian Monte Carlo*, **Winner Outstanding Poster Award**)
Masroor Khonkhodzhaev (*Dynamic Programming and Reinforcement Learning*)
Percy Martinez (*Fourier Analysis*)
Isabella Chittumuri (*Elements of Statistical Learning*)
- 2023 Stony Brook DRP
Shailen Smith (*Probabilistic Machine Learning*)
- 2022 Bayesian Data Analysis for the Global South
Volunteer TA for online course by Aki Vehtari [\[certificate\]](#)

Updated January 2026